

### Lab 04-1: Verdi

Assigned on Sep 25, 2025

Due day on Oct 02, 2025

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## Objective

In this lab, you will learn:

1. A simple CPU includes instruction set architecture, dataflow modeling and controller design.
2. Use Verdi to trace and debug legacy RTL codes.

Demo checklist:

- ☐ Show TA your simulation result of fixed CPU.

## Environment Setup

Copy lab file packages from ee4292. Decompress the package and enter it. You can check the file list in Appendix.

```
$ cp ~ee4292/iclab2025/lab04.zip .
```

```
$ unzip lab04.zip
```

```
$ cd lab04/lab4_part1/
```

Note: please run simulation in the *sim* directory to maintain a fine data management.

## Description

A simple but buggy CPU in RTL is provided in this lab. In these two weeks, we will adapt this CPU and add some functions. But first, you need to trace the legacy verilog code and to understand how the CPU is implemented. As introduced in class, tool *Verdi* can help you trace unfamiliar codes and debug, so finish the lab with Verdi and provided document CPU.

## Action Items

### I. Use Verdi to debug legacy codes.

1. Use nSchematic to observe the overall CPU.
2. There are some bugs in the CPU, use nWave or nSchematic and the provided testbench to fix these bugs. You will see “*Congradulation! All of the registers are correct.*” if all bugs are cleared.

## Appendix

| Directory | Filename            | Description                        |
|-----------|---------------------|------------------------------------|
| sim       | run.f               | Simulation file list               |
| sim       | hdl.f               | Pure HDL file list                 |
| sim       | test_top.f          | Test bench for this CPU            |
| sim       | golden_register.txt | true gpr state of CPU              |
| sim       | golden_register.pat | true gpr state of CPU(hex pattern) |
| sim       | instruction.txt     | CPU simulation pattern             |
| source    | top.v               | Top module for this CPU            |
| source    | ID_stage.v          | RTL for ID stage                   |
| source    | controller.v        | RTL for control signal             |
| source    | regfile.v           | RTL for register file              |
| source    | ID_EXE.v            | RTL for ID EXE flip flop           |
| source    | EXE_stage.v         | RTL for EXE stage                  |
| source    | alu.v               | RTL for alu                        |
| source    | CPU_define.v        | Define file                        |